

MQAT: Module 2 - Database, Exploratory Data Analysis (EDA)

Exercises in R

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Table of contents

1	Quantitative Methods of Analysis in Transportation course	1
	Introduction to R programming	3
	Why R and GIS	3
	Recommended readings	3
2	Software	5
2.1	R	5
2.1.1	Windows	5
2.1.2	Mac	5
2.1.3	Ubuntu	6
2.2	IDE	7
2.2.1	Positron	7
2.2.2	Windows	7
2.2.3	MacOS	8
2.2.4	Ubuntu	8
2.2.5	RStudio	8
2.2.6	Windows	8
2.2.7	MacOS	8
2.2.8	Ubuntu	8
2.3	R packages	8
2.4	Cheat sheets	10
	References	13

1 Quantitative Methods of Analysis in Transportation course

Materials, exercises, data and tutorials for the [Quantitative Methods of Analysis in Transportation](#) course of the [MSc in Transportation Systems](#) at Instituto Superior Técnico - University of Lisbon.



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🔥 This is not the official book for the course.

Please refer to Fénix: [Quantitative Methods of Analysis in Transportation](#)

Introduction to R programming

This book aims to provide tools to deal with transportation datasets using [R programming](#), an open-source and widely used tool for data analytics in urban mobility.

It includes course sporting materials, exercises, and examples to explore with **R**, with a focus on modelling and spatial data analysis using GIS techniques.

Why R and GIS

Most academic programs focus on teaching modelling and deep analysis of data. However, there is a need to learn how to explore and prepare a dataset for modelling. The use of **programming and GIS** techniques have enormous advantages, including their flexibility; reproducibility; and transparency and understanding the step-by-step process.

The use of GIS techniques in transportation is of enormous relevance when doing accessibility analysis or reeling with georeferenced transportation data, such as bike sharing route trips' datasets, origin-destination flows datasets, home/work locations, GTFS public transit data, and so on. There is a need to learn how to locate these open datasets, how to explore them and how to integrate them into transportation and urban analysis. Additionally, the use of open source software and datasets allows researchers to perform methods that are reproducible and transparent.

TLDR

- Open-source tools widely used in data analytics and spatial analysis
- Flexibility and reproducibility in data manipulation and visualization
- Critical for urban mobility and transportation research, with spatial relevance
- Large transportation datasets are becoming increasingly common

Recommended readings

- Engel (2023) [Introduction to R](#).
- Wickham et al. (2017) [R for Data Science](#).
- Lovelace et al. (2024) [Geocomputation with R](#).

2 Software

In this chapter we will guide you through the installation of R, Positron/RStudio IDE and the packages you will need for this course.

R and **Positron/RStudio**¹ are separate downloads.

2.1 R

You will need **R** installed on your computer. **R stats** (how it is also known) is a programming language and free software environment for statistical computing and graphics supported by the R Foundation for Statistical Computing.

The download links live at [The Comprehensive R Archive Network](#) (aka CRAN). The most recent version is 4.6.1, but you can use any $\geq 4.1.x$ if you already have it installed.

2.1.1 Windows

[Download R for Windows](#) and run the executable file.

Important

You may also need to install [Rtools](#), which is a collection of tools necessary to build R packages in Windows.

2.1.2 Mac

[Download R for MacOX](#). You will have to choose between the *arm64* or the *x86-64* version.

Download the `.pkg` file and install it as usual.

¹We will use Positron. Although if you prefer RStudio or already use other IDE such as VScode, that's also fine.

2.1.3 Ubuntu

! These are instructions for Ubuntu. If you use other linux distribution, please follow the instructions on [The Comprehensive R Archive Network - CRAN](#).

You can look for R in the Ubuntu **Software Center** or install it via the terminal:

```
# sudo apt update && sudo apt upgrade -y
sudo apt install r-base
```

Or, if you prefer, you can install the latest version of R from CRAN:

```
# update indices
sudo apt update -qq
# install two helper packages we need
sudo apt install --no-install-recommends software-properties-common
dirmngr
# add the signing key (by Michael Rutter) for these repos
wget -qO- https://cloud.r-project.org/bin/linux/ubuntu/marutter_pubkey.asc
| sudo tee -a /etc/apt/trusted.gpg.d/cran_ubuntu_key.asc
# add the R 4.0 repo from CRAN
sudo add-apt-repository "deb https://cloud.r-project.org/bin/linux/ubuntu
$(lsb_release -cs)-cran40/"
```

! Using an Ubuntu based distribution, but not pure Ubuntu?

If you are using an Ubuntu based Linux distribution, such as Linux Mint, the last command from the previous script will set the CRAN repository to an invalid address, causing the next step to throw an error when trying to fetch and install R.

To fix it, run the commands below.

```
# Remove invalid repository defined before
sudo add-apt-repository --remove "deb
https://cloud.r-project.org/bin/linux/ubuntu $(lsb_release
-cs)-cran40/"
# Set repository to fixed address, considering Ubuntu version
(upstream) instead of your Ubuntu flavour one (which is invalid
for CRAN)
sudo add-apt-repository "deb
https://cloud.r-project.org/bin/linux/ubuntu $(source
/etc/upstream-release/lsb-release && echo
$DISTRIB_CODENAME)-cran40/"
```

Then run:

```
sudo apt install r-base r-base-core r-recommended r-base-dev
```

Optional

To keep up-to-date R version and packages, you can follow the instructions at [r2u](https://r2u.github.io).

After this installation, you don't need to open R base. Please proceed to install the IDE.

2.2 IDE

An IDE (Integrated Development Environment) is a software that provides tools to write, test and execute code in a single graphical interface. It makes the process of programming easier and more accessible to users less proficient in coding.

Below are tutorials to install two alternatives that support R. Both include a console, syntax-highlighting editor that supports direct code execution, as well as tools for plotting, history, debugging and workspace management. Positron has a more modern UI, but RStudio is more established due to its longevity. You can choose either.

2.2.1 Positron

Positron is available for free download from [Posit Positron](https://positron.posit.co/).

2.2.2 Windows

[Download Positron](#) for Windows and run the executable file.

User or system install?

You can choose either. User install will keep Positron available only for your user, while System install will make it available to any user on your computer.

Is Positron failing to start R session?

If after successfully installing Positron you open it and face an error on the R console, installing Microsoft Visual C++ Redistributable can resolve missing dependencies that might be causing the problem.

To install it, open [this link](#) and download the `.exe` file in the **Link** column that suits your system architecture. Click on the file and run the installation procedure. Restart Positron. The issue should be gone.

2.2.3 MacOS

[Download Positron](#) for macOS file and install it as usual.

2.2.4 Ubuntu

[Download Positron](#) .deb file and install it as usual.

2.2.5 RStudio

RStudio is available for free download from [Posit RStudio](#).

2.2.6 Windows

[Download RStudio IDE](#) .exe file and run the executable file.

2.2.7 MacOS

[Download RStudio IDE](#) .dmg file and install it as usual.

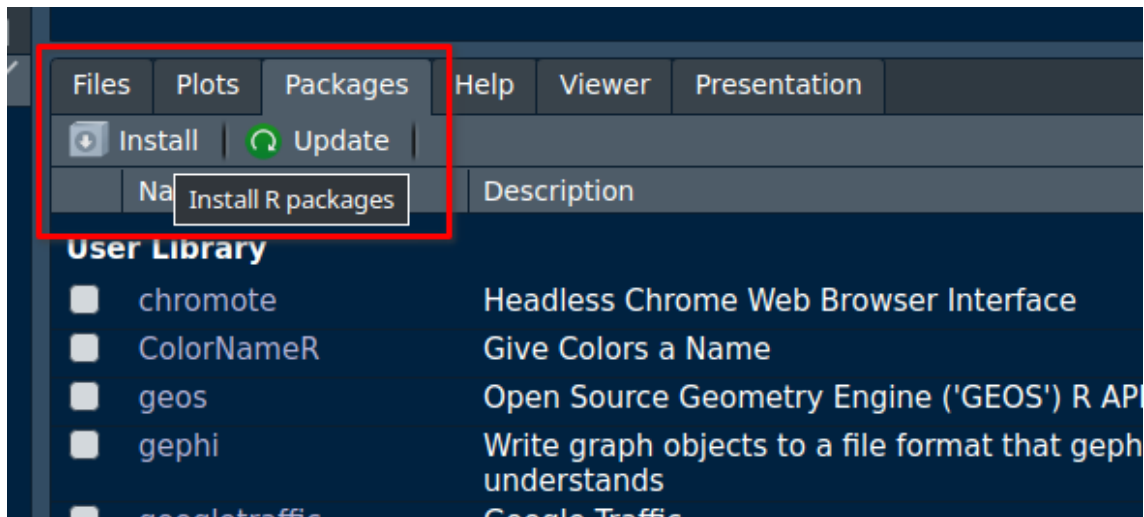
2.2.8 Ubuntu

[Download RStudio IDE](#) .deb file and install it as usual.

2.3 R packages

You will need to install some packages to work with the data and scripts in this course.

You can install them in RStudio by searching for them in the **Packages** tab:



or by running the following code in the console:

```
install.packages("tidyverse")
install.packages("readxl")

install.packages("sf")
install.packages("mapview")
install.packages("rmarkdown")
install.packages("centr")
install.packages("od")
install.packages("openrouteservice")

install.packages(c("remotes", "devtools", "usethis")) # optional
install.packages("osmextract") # optional
install.packages("stplanr") # optional
```

⚠️ Facing problems installing packages on Linux?

It is possible that the installation of the packages above fails due to missing dependencies. Unlike Windows or Mac, Linux sometimes tries to “build” packages from scratch, which causes errors if certain system tools are missing. To avoid this, follow the steps below to tell R to download ready-to-use packages instead of trying to build them.

1. On the **Terminal** (next to the console, at the bottom of your screen, there is a tab called **Terminal**), run the command below to get your OS **codename**:

```
# If you are using an Ubuntu original distro, get the value at
# 'codename'
lsb_release -a
# If you use other Ubuntu based distro (such as Linux Mint), get
# the value returned by the command below
source /etc/upstream-release/lsb-release && echo $DISTRIB_CODENAME
```

2. Back on the **Console**, run the following R code, making sure to replace ‘resolute’ with the codename retrieved in the previous step.

```
# Use pre-compiled binary packages to avoid errors due to missing
# dependencies
options(
  repos = c(CRAN =
    "https://packagemanager.posit.co/cran/__linux__/resolute/latest"),
  HTTPUserAgent = sprintf("R/%s R (%s)", getRversion(),
    paste(getRversion(), R.version["platform"],
    R.version["arch"], R.version["os"])))
)

# Disable bspm interception (use pre-compiled Linux binaries
# directly from Posit without needing apt or system privileges)
# install.packages(c("bspm"))
bspm::disable()
```

3. All set! Try to install the packages again. It should run smoothly now. :)

2.4 Cheat sheets

i This is the only cheating allowed at this course.

Cheatsheets are a concise set of notes used for quick reference. Posit, the open-source data

science software corporation responsible for RStudio/Positron IDEs, also makes available several cheat sheets that provide a comprehensive introduction to many of R softwares and packages.

You can start by taking a look at the ones for [Positron](#) or [RStudio](#).

Refer to rstudio.github.io/cheatsheets for the full cheat sheet repository.

References

Engel, Claudia A. 2023. *Introduction to r*. cengel.github.io/R-intro/.

Lovelace, Robin, Jakub Nowosad, and Jannes Muenchow. 2024. *Geocomputation with r*. 2nd ed. Chapman; Hall/CRC. <https://r.geocompx.org/>.

Wickham, Hadley, Mine Çetinaka-Rundel, and Garrett Grolmund. 2017. *R for Data Science*. 2nd ed. O'Reilly Sebastopol. <https://r4ds.hadley.nz/>.

